Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/26/2005
PI Name:	Maxworthy, Tony Ph.D.		
Project Title:	The Dynamics of Miscible Interfaces: A Space Flight Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	213 740-0481
Organization Name:	University of Southern California		
PI Address 1:	Department of Aerospace and Mechanical Engineering		
PI Address 2:			
PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90089-1453	**Congressional District:**	33
Comments:	9/7/16 - PI is deceased (March 8, 2013) per online university obituary.		
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	01/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Balasubramaniam, R.	**Contact Phone:**	216-433-2878
Contact Email:	RBalasubramaniam@grc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2006 (from 12/31/2006) per PI information in 2007 (1/2009) Note: grant renewed 4 yrs		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GB29G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Both theoretical and experimental research will be performed. In the experimental studies, a miscible liquid displaces another liquid in a cylindrical tube. The displacement velocity is such that the Peclet number for the mass transfer is large compared to unity. The velocity and concentration fields near the mixing front are obtained by particle image velocimetry and interferometry respectively. The experimental results will be compared in-depth with computational results that include Korteweg stresses and the non-vanishing divergence of the velocity field in the mixing region. The computations will predict the spatio-temporal evolution of the velocity and concentration fields for a variety of system parameters. The objective of this research is to try to unravel a number of difficult issues that are present in the study of miscible interfaces. It is known that in addition to the usual stresses in a fluid, additional stresses called Korteweg stresses could be important in regions where the concentration gradients are large. Also, in the mixing region, the velocity field is not divergence free in incompressible fluids. The goal of this research is obtain the flow and concentration fields near miscible interfaces in microgravity and compare them with computational results that include the Korteweg stresses and divergence. The improved understanding of the dynamics of mixing that will be obtained from this research can be applied to multiphase flows in porous media such as those occurring in enhanced oil recovery, fixed bed regeneration, hydrology and filtration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Following the works of year before, we found some new and interesting problems regarding the finger displacement in capillary. One is to quantify the displaced fluid left on the tube wall. The fraction m was used as a parameter for this but our research found this parameter itself does not represent the thickness of displaced fluid on the tube wall. The second problem we focused is the interfacial stability. Experiments regarding this problem were conducted using immiscible fluids in capillary tubes.
Bibliography Type:	Description: (Last Updated: 01/26/2009)
Articles in Peer-reviewed Journals	Kuang J, Petitjeans P, Maxworthy T. "Velocity fields and streamline patterns of miscible displacements in cylindrical tubes." Experiments in Fluids 2004 Aug;37(2):301-8. http://dx.doi.org/10.1007/s00348-004-0824-0 , Aug-2004